A Clinical Workflow Integration Layer for Tuberculosis Chest X-ray Screening

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***Abstract*—Tuberculosis (TB) causes more deaths than any other infectious disease worldwide. However, high-burden countries are often short-staffed on radiologists to interpret chest X-rays for TB screening. Hence, computer-aided detection (CAD) systems for TB screening are in wide use. While TB detection models are highly accurate, several practical challenges repeatedly emerge when integrating these systems into clinical workflows. Commercial TB CAD systems often do not account for X-ray image quality, prioritization of abnormalities found, post-deployment monitoring, and clinical explanation of findings for interpreting radiologists. We present an academic prototype of a clinical workflow integration layer to address these issues. Its lead contribution is a clinically grounded LLM explanation module that describes model findings in natural language for radiologists, a feature not yet present in the commercial products we reviewed. Supporting modules add an input quality gate to assess X-ray image quality, a structured evidence interface to present findings, and a post-deployment monitoring dashboard. Alongside the workflow layer, we also develop and benchmark custom models for tuberculosis classification and localization. Our results show that ... (TODO).**

***Keywords*—Tuberculosis; Computer-Aided Detection; Chest X-ray; Clinical Workflow; Medical Imaging; Computer Vision**

I. Introduction

Tuberculosis (TB) is an airborne bacterial infection caused by *Mycobacterium tuberculosis* that primarily affects the lungs. It spreads when an infected person coughs, sneezes, or speaks, releasing bacteria that others may inhale [3]. TB is both preventable and curable, yet it causes more deaths than any other infectious disease worldwide [1]. Low- and middle-income countries bear a disproportionate share of this burden, often compounded by limited access to diagnostics in rural areas, delayed presentation, and overburdened public health infrastructure [2].

Chest X-ray (CXR) is the most widely used imaging modality for TB screening [4]. In screening settings, CXR serves as a triage tool. Patients whose CXR show findings suggestive of TB are referred for bacteriological testing to confirm the presence of TB, typically through sputum smear microscopy or molecular assays. This two-stage workflow of TB CXR triage followed by laboratory confirmation is central to national TB control programs in high-burden countries [2].

High-burden countries are often short-staffed on radiologists to interpret CXRs for TB screening. This scarcity is especially of concern in primary care facilities and rural health units where screening volumes are highest [2]. This bottleneck has driven the development of computer-aided detection (CAD) systems that use medical imaging models to analyze CXR images and flag those suggestive of TB.

Radiologist performance in TB screening has been demonstrated to agree with or improve with AI assistance. A 2024 reader study demonstrated that AI assistance raised physician diagnostic performance, measured by the area under the receiver operating characteristic curve, from 0.773 to 0.874. Non-radiologist physicians with AI assistance were also able to achieve results on par with trained radiologists [10]. In a real-world observational study, radiologists showed complete agreement with AI findings in 86.5% of cases, with only 0.5% of findings missed by the model [11].

A. Maturity of TB CAD Systems

TB CAD systems have been endorsed by the World Health Organization (WHO). In 2021, WHO issued a policy recommendation supporting CAD as an alternative to human readers for TB screening [6]. In 2025, the WHO Technical Advisory Group approved six commercial CAD products as meeting its performance standards for TB screening in individuals aged 15 years and older [7].

Model performance of TB CAD systems have achieved diagnostic accuracy in range with WHO screening targets. A systematic review of machine learning for chest X-ray interpretation found strong diagnostic performance across a range of thoracic conditions, though it also noted limited prospective validation and real-world deployment evidence [9]. Several other studies find CAD systems approach WHO targets [40] [41]. These indicate that the field has moved beyond achieving diagnostic accuracy into integrating TB CAD systems in routine clinical workflows.

B. Gaps in Clinical Workflow Integration

Despite strong detection performance, reviews of CXR AI consistently identify gaps between laboratory accuracy and clinical implementation. A review of practical AI in chest radiography found that only a small proportion of published studies are prospective or clinically embedded, and that workflow integration, local validation, and human factors remain under-studied [12]. Hence, clinical expertise, usability, and implementation context remain essential even when model performance is strong [9], [12]. Real-world deployment studies demonstrate this gap as well. An implementation of qXR for TB screening in India identified barriers including software installation, workflow integration, network connectivity, and limited human resources for interpreting findings [37]. Another deployment study in Vietnam found only 47.3%

TABLE I. FEATURE COMPARISON WITH COMMERCIAL TB CAD PRODUCTS

Feature	**qXR** [21]	**CAD4TB+** [22]	**Lunit** [23]	**DrAid** [25]	**Genki** [26]	**InferRead** [27]	**Siemens** [24]	**Proposed**
Chest X-ray quality assessment				*n/a*	*n/a*		✓	✓
Monitoring and analytics dashboard			✓	*n/a*	*n/a*			✓
Heatmap/overlay of localization findings	✓	✓	✓	✓	✓	✓	✓	✓
Explanation of findings in natural language				*n/a*	*n/a*			✓

✓ marks a feature documented as present; blank marks a feature documented as absent or not offered; *n/a* marks features for which no documentation was found.
The monitoring/analytics dashboard column denotes a post-deployment dashboard for tracking model usage and performance over time.

concordance between physician and CAD interpretations at three months [38], and early user interviews across multiple countries found that clinicians valued CAD most when it complemented rather than replaced clinical judgment [39]. Usability testing of CXR AI in a simulated clinical workflow reported that users wanted clearer explanations of model findings and suggested the tool would be most useful for understaffed radiology departments, trainees, and non-radiologist physicians [13]. Several gaps emerge from the literature:

CXR image quality varies in real-world use. CAD systems are trained on curated datasets. In contrast, in routine hospital use, images may be poorly positioned, clipped, rotated, underexposed, or photographed from analog film. Poor input quality can degrade performance, but most systems do not expose a user-facing quality assessment before inference [12], [14].

Heatmaps for explaining model findings lack clinical utility. All of the commercial CAD products reviewed in Table I provide heatmap overlays showing image regions that influenced the prediction. However, radiologist user studies report that heatmap-style saliency overlays are less intuitive and harder to read than localized, discrete findings [43]. A systematic review also found that saliency-based approaches often show low overlap with expert-identified regions of interest and do not convey the contextual reasoning clinicians use during diagnosis [29]. Alternative approaches, including textual descriptions and case-based reasoning, have been proposed as more clinically meaningful [30] [15].

Post-deployment monitoring is often lacking. A deployed system's performance may change due to population shifts, equipment changes, or software updates. The U.S. Food and Drug Administration (FDA) has identified data drift, concept drift, and model drift as risks to AI-enabled medical devices and has initiated research programs on proactive post-market monitoring [33]. Reviews note the need for ongoing monitoring, but this remains an operational gap in practice [12].

C. Related Work

Academic prototypes have begun coupling detectors or image encoders with language models to provide natural language explanation of findings. Work from Siemens Healthineers, who also developed one of the commercial TB CAD systems compared against in this paper, couple a thoracic abnormality detector with a domain-adapted large language model to generate chest-X-ray findings [15], although this functionality is not yet implemented in the corresponding commercial system. General medical vision-language models such as LLaVA-Med [16] and CXR-LLaVA [31] demonstrate image-conditioned radiology text generation, and recent surveys catalogue the rapid growth of vision-language models for medical report generation and visual question answering [32]. These remain research prototypes rather than features in deployed commercial systems.

We reviewed seven commercial CAD products to compare features addressing common gaps in clinical workflow integration. Six of these are approved by WHO as meeting its performance standards for TB screening : qXR v4.0 (Qure.ai), CAD4TB v7 (Delft Imaging), Lunit INSIGHT CXR v3.1, DrAid for TB screening v2.4.4–6 (VinBrain), Genki Edge v3.4.2 (Deeptek), and InferRead DR Chest v1.0.1.1 (Infervision)[7]. We additionally compare Siemens AI-Rad Companion, which has done relevant work on the gaps mentioned in this paper. Specifically, Siemens implements input-quality assessment, and has conducted research into language-based explanation of findings as discussed above (although the latter is not implemented in the commercial product).

As Table I summarizes, all products provide heatmap or overlay-based localization. However post-deployment monitoring is uncommon, input quality assessment is largely absent, and no product is known to explain model findings as natural-language clinical reasoning.

D. Scope and Contribution

This paper addresses gaps in integration into clinical workflows by presenting a prototype clinical workflow integration layer for TB CAD screening in CXRs. The contributions of this system are:

1) A **natural language explanation module** that generates natural-language descriptions of AI-detected findings, linking suspected abnormalities to localized visual evidence. This is the lead contribution.
2) An **input quality assessment module** that screens radiographs for acquisition defects before inference.
3) A **structured evidence interface** presenting the TB score, risk classification, quality status, and localized evidence in a compact clinical panel.
4) A **post-deployment monitoring dashboard** tracking system behavior over time.

As a secondary contribution, we develop and benchmark several custom models for image-level classification and bounding-box localization.

II. SYSTEM ARCHITECTURE

The system processes a CXR through a sequential pipeline:

1) **Image ingestion.** A digital chest radiograph (DICOM or standard image format) is received from the hospital imaging system or uploaded directly.

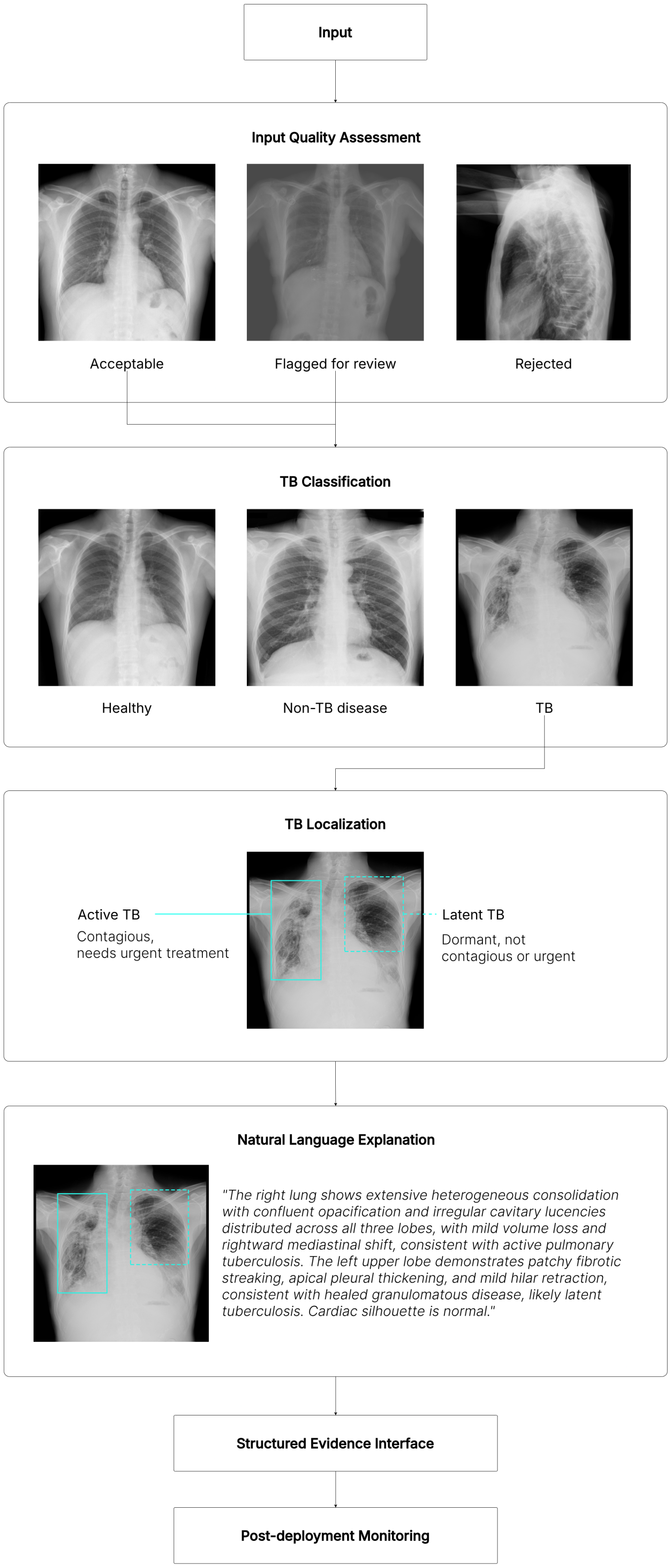

Fig. 1. Overview of the proposed TB CAD pipeline.

2) **Input quality assessment.** The image is evaluated for acquisition quality before diagnostic inference.
3) **TB classification.** If the image passes quality screening, it is processed by the TB scoring module, producing a continuous probability score and spatial activation maps.
4) **Multimodal explanation generation.** For classified cases, the system generates a natural-language explanation linking the predicted finding to localized visual evidence.
5) **Clinical interface rendering.** All outputs are presented in a structured evidence panel.
6) **Monitoring and logging.** Each case and its metadata are logged for the monitoring dashboard.

The following sections describe each component: input quality assessment (Section III), tuberculosis classification (Section IV), tuberculosis localization (Section V), and the multimodal explanation module (Section VI), followed by two non-evaluated components, the structured evidence interface (Section VII) and the post-deployment monitoring dashboard (Section VIII).

III. Input Quality Assessment

IV. Tuberculosis Classification

The TB classification module classifies each CXR as either healthy, sick non-TB, or TB as seen in Figure 1.

A. Dataset

Classification and localization models are trained on TBX11K, a public dataset of 11,200 CXRs with image-level labels (*healthy*, *sick non-TB*, *TB*) for classification and bounding box-level labels (*active-TB* and *latent-TB*) [49] for localization (Section V). We use a redistribution of TBX11K with labels simplified for usability[1].

TBX11K was selected as it is one of the few publicly available TB CXR datasets that satisfies the following:

- Large TB-specific split to train networks from scratch.
- Expert bounding-box annotations for TB regions.
- Demographic composition and class imbalance which reflect real-world clinical settings [50]. The sick non-TB class is important to this realism, since in deployment a radiologist receives scans from the full patient stream, many of whom are sick with conditions other than TB, so a clinically useful model must separate TB from other disease.

The official TBX11K test set is withheld for an online challenge, so we use the publicly available validation set as our test set for all experiments. Our results should therefore not be compared directly to those reported for SymFormer [50] or other entries on the online challenge, which are measured on the private test set.

B. Data Preprocessing

Classification models are trained without preprocessing or data augmentation since low-quality inputs are already filtered out by the input quality gate (Section III). In addition, CXRs are acquired under a standardized posteroanterior protocol and are largely uniform in pose, so aggressive augmentation risks introducing distribution shift.

[1]https://www.kaggle.com/datasets/vbookshelf/tbx11k-simplified

C. Model Architecture

We evaluate nine classifiers: six standard baselines and three custom models. Baselines comprise EfficientNet-B0 [61], MobileNetV3-Large [60], DenseNet-121 [59], ResNet-18, ResNet-50 [58], and ConvNeXt-Tiny [62]. Custom models are DraxNet, Drax-MobileNetV3-Large, and FlipR.

1) DraxNet: DraxNet is a custom model (17.0M parameters) based on a ResNet-18 backbone. The main modification is that the fourth layer consists of two ResNet blocks, each with a custom mixer (*DraxBlock*) inserted between its two convolutions. All other components are unchanged from ResNet-18.

DraxBlock is a conv-attention hybrid that consists of a ConvNeXt local convolutional branch followed by an optional self-attention branch. Residuals are fused through a stochastic-depth block.

DraxNet refers to Dynamic Residual Attention eXchange. The rationale is to use the well-understood ResNet-18 backbone for the high-resolution early stages and introduce the heavier custom mixer only at the final, lowest-resolution stage where the feature map is most compact, which adds attention capacity with fewer additional parameters and compute.

2) Drax-MobileNetV3-Large: Drax-MobileNetV3-Large is a second custom variant (6.1M parameters) that applies the same DraxBlock idea to a more parameter-efficient backbone. The MobileNetV3-Large feature extractor and classification head are kept unchanged, and a bottlenecked *DraxBlock* is inserted after the final feature stage, keeping the overall model close to MobileNetV3-Large in parameter count while testing whether the DraxBlock generalizes beyond a ResNet-style backbone.

3) FlipR: FlipR is a lightweight custom model (2.8M parameters) built on a ResNet-18 backbone. The ResNet-18 stem, layers, and classification head are unchanged. A custom block, named *FlipR*, is inserted between layers.

The *FlipR* block computes the difference between each feature map and its horizontally flipped counterpart after a smoothing step with average pooling, which accounts for imperfect anatomical symmetry. From this, a gated mask is broadcast across all channels at once.

The block is inspired by SymAttention [50], which observes that bilateral asymmetry in CXRs is a key indicator of TB. FlipR tests whether a lightweight symmetry gate can achieve this over heavier attention-based approaches.

D. Model Training

All models are trained from scratch with a fixed random seed, batch size of 16, 512×512 input resolution, and 100 epochs, using Adam with a fixed learning rate of 10^{-3}, no learning-rate schedule, and an unweighted cross-entropy loss. No hyperparameter tuning was done.

E. Results

Classification results are reported in Table II.

MobileNetV3-Large is the top classifier, while FlipR achieves comparable results with half of its parameters. Due to the single-seed, single-split evaluation and small ($\pm 3\%$) differences in performance, the ordering among the top four (FlipR, EfficientNet, MobileNetV3-Large, Drax-MobileNetV3-Large) can likely be attributed to stochasticity. The four differ most on sensitivity ($\pm 2.5\%$ difference), with MobileNetV3-Large (96.67% sensitivity) interpreted as the top model since sensitivity is more clinically important in TB screening. Despite lower sensitivity, FlipR achieves comparable results on other metrics with roughly half the size (2.8M vs 5.5M for MobileNetV3-Large). FlipR's results support the hypothesis from SymFormer [50] that bilateral asymmetry is highly indicative of TB.

All models meet WHO targets for TB triage. Sensitivity ranges from $92.50 - 96.67\%$, meeting the WHO target of minimum 90% sensitivity. Specificity also ranges from $97.98 - 99.91\%$, 25% above the WHO target of minimum 70% specificity [8]. Models are more likely to classify a CXR as no TB or sick non-TB, and rarely raise false alarms. For TB screening, sensitivity is more clinically important than specificity, since a missed TB (false negative) case results in untreated TB and risk of transmission, whereas false positives result in unnecessary confirmatory testing. The top model, MobileNetV3-Large, misses only 3.3% of TB cases.

High specificity, lower sensitivity is likely attributable to the TBX11K dataset rather than model architecture. Related literature shows that various models' performance on TBX11K consistently achieve similar results of high specificity and lower sensitivity. The TBX11K dataset frames high specificity as the desirable outcome on the dataset, since the common failure mode of models on other TB datasets is over-predicting TB and achieving near-zero specificity [50].

Lightweight models are preferred for TB screening for both performance and practical use. Smaller models outperform larger ones across the board. MobileNetV3-Large (5.5M), EfficientNet-B0 (5.3M), and FlipR (2.8M) all achieve higher accuracy and F1 than larger models. The TBX11K training set is modest and the TB class is the minority, with no techniques applied for class imbalance, so the additional capacity of larger networks appears to overfit the majority healthy and sick non-TB classes rather than learning discriminative TB features. TB screening is also concentrated in high-burden, resource-limited settings where inference frequently runs without a dedicated GPU, so a lightweight model is preferable in clinical settings.

V. Tuberculosis Localization

A. Dataset

Localization uses the bounding-box-level annotations of the TBX11K dataset, where each region is classified as either active TB or latent TB [49].

Active TB denotes ongoing infection that is transmissible and requires treatment. Latent TB denotes prior, healed or inactive infection that is not contagious and does not require immediate treatment, although may be at risk of reactivation. As seen in Figure 1, active TB is more immediately visible, and latent TB is more subtle.

A CXR classified as TB at the image-level may contain only active-TB lesions (924 images), only latent-TB lesions (212 images), or lesions of both types (54 images), as depicted

TABLE II. TUBERCULOSIS CLASSIFICATION RESULTS

Model	AUC_{TB}	Acc. (%)	Sens. (%)	Spec. (%)	Avg. Prec. (%)	Avg. Rec. (%)	Avg. F1 (%)	Params
FlipR	0.9984	98.97	94.17	**99.91**	**99.01**	97.70	98.34	**2.8M**
EfficientNet-B0	**0.9995**	98.89	95.00	99.65	98.29	97.87	98.07	5.3M
MobileNetV3-Large	**0.9995**	**99.21**	**96.67**	99.82	98.97	**98.54**	**98.75**	5.5M
Drax-MobileNetV3-Large	0.9989	98.81	95.00	99.65	98.23	97.81	98.02	6.1M
DenseNet-121	0.9975	98.57	92.50	99.56	97.81	96.97	97.38	8.0M
ResNet-18	0.9982	98.17	93.33	99.21	96.71	96.90	96.80	11.7M
DraxNet	0.9988	98.41	95.00	99.47	97.51	97.51	97.51	17.0M
ResNet-50	0.9978	97.70	92.50	99.47	96.95	96.33	96.63	25.6M
ConvNeXt-Tiny	0.9952	97.14	95.00	97.98	93.63	96.58	94.97	28.6M

AUC_{TB} is the area under the ROC curve for the TB class. Sensitivity is recall for the TB class. Specificity is recall for the non-TB class (healthy and sick non-TB combined). Precision, recall, and F1 are averaged across the three classes (healthy, sick non-TB, TB).

in Figure 1. This class composition is consistent with real-world conditions.

B. Model Architecture

We evaluate YOLO26 and a custom variant, YOLO26 with a DraxNet backbone. In the custom model, the detector's default feature extractor is replaced by the DraxNet backbone used in the classification experiments. This isolates the effect of the backbone on TB localization while the detection head and training protocol are unchanged.

C. Model Training

Localization uses the same configuration as model training for classification models reported in Section IV.

TABLE III. TUBERCULOSIS LOCALIZATION RESULTS

Model	Class	mAP_{50}	$\text{mAP}_{50\text{-}95}$
YOLO26	ATB	—	0.2110
YOLO26	LTB	—	**0.0410**
YOLO26 + DraxNet	ATB	—	**0.2748**
YOLO26 + DraxNet	LTB	—	0.0310

ATB = active TB; LTB = latent TB.

D. Results

Localization results per class are reported in Table III.

Active TB is more likely to be detected than latent TB. The very low $\text{mAP}_{50\text{-}95}$ for latent TB is likely due to the class imbalance in TBX11K (active-TB regions outnumber latent-TB regions by roughly 4:1) and the subtle appearance of latent-TB lesions. The TBX11K reference likewise reports all detection methods struggle with latent TB, attributing this to the class imbalance [50]. This is also consistent with human reader performance, which distinguish latent TB less reliably than active-TB findings [5]. Since active TB requires urgent treatment as compared to latent TB, this therefore aligns with clinical triage priorities.

The DraxNet backbone specializes toward active TB. YOLO26 with DraxNet backbone improves active TB localization substantially over vanilla YOLO26, raising $\text{mAP}_{50\text{-}95}$ from 0.2110 to 0.2748 (+0.0638), but slightly reduces LTB from 0.0410 to 0.0310. This gain is driven almost entirely by precision, which rises from 0.74 to 0.82, while recall is unchanged at approximately 0.26.

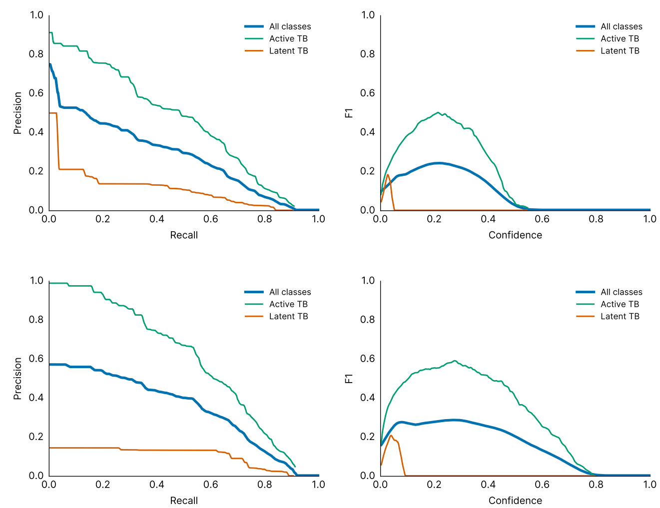

Fig. 2. Localization precision-recall (left) and F1-confidence (right) curves for YOLO26 (top) and YOLO26 with the DraxNet backbone (bottom). Both sustain high precision while recall saturates near 0.26, and the latent-TB curve collapses at low confidence, so localization functions as a high-precision rule-in cue rather than a standalone detector.

Recall, not precision, is the bottleneck. Both detectors localize confidently but detect only about a quarter of the annotated TB regions. In Figure 2, precision stays high while recall saturates near 0.26, and the latent-TB curve collapses at low confidence for both detectors.

Localization is useful to detect TB, but should not be used to exclude TB. Clinically, the high-precision, low-recall profile defines how localization should be used. A flagged lesion is relatively likely to be correct and thus warrants focused radiologist review, but the detector must not be used to exclude TB since most annotated lesions are missed. This indicates that localization is most reliable as supporting region-level evidence for the classifier and LLM explanation modules, rather than as a standalone detector.

E. Implementation Notes

Localization runs only on CXRs the classifier (Section IV) flags as TB. CXRs ruled non-TB skip localization, since any lesions found would contradict the classifier's decision.

VI. Multimodal Explanation

VII. Structured Evidence Interface

VIII. Post-Deployment Monitoring

IX. Discussion

A. Limitations

B. Future Work

X. Conclusion

Declaration on Generative AI

Generative AI tools were used to assist in drafting and preparing this manuscript. The conception of the research, methodology, and conclusions are the work of the authors.

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